

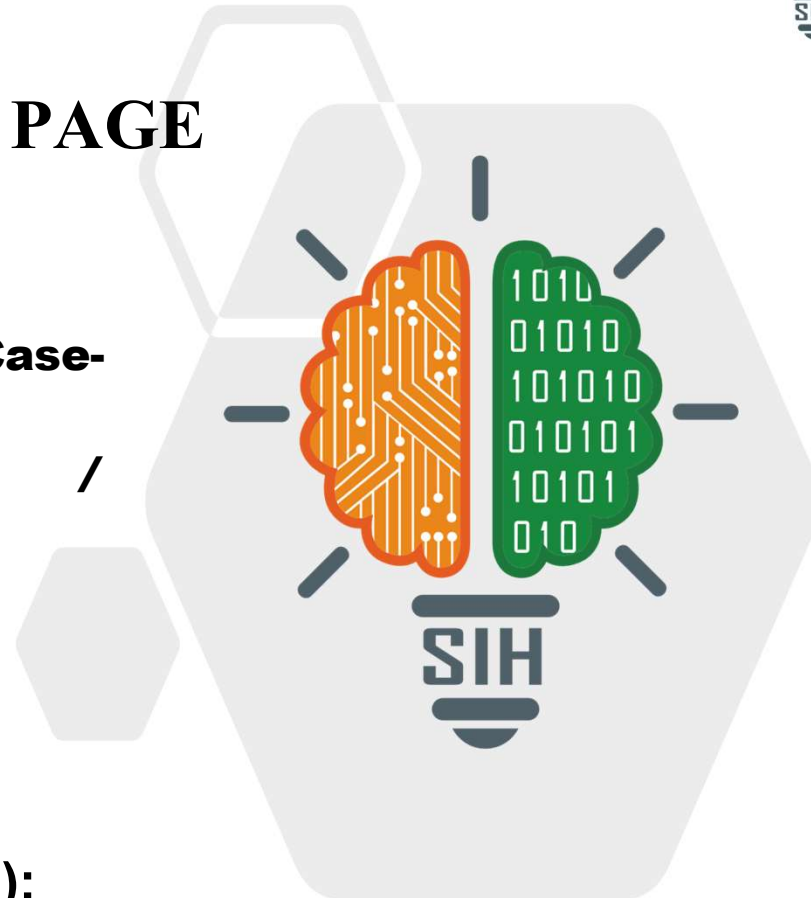
SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

TITLE PAGE

- Problem Statement ID –26047
- Problem Statement Title-**Patient Case-Taking Software**
- Theme-**MedTech** / **BioTech** / **HealthTech**
- PS Category- Software
- Team ID-
- Team Name (Registered on portal):



IDEA TITLE

MEDIKIOSK – PROPOSED SOLUTION

- **Detailed Explanation**

AI-powered kiosk using **Voice, Touch & OCR** to collect patient history and generate structured clinical summaries.

- **How It Addresses the Problem**

Reduces **doctor workload, consultation time and repeated questioning** through pre-consultation data collection.

- **Innovation & Uniqueness**

Combines **Multilingual AI, Adaptive Questioning, OCR, AYUSH History, Red-Flag Detection & Doctor Validation** in one platform.

Key Value: Faster • Smarter • Multilingual • Patient-Centric Healthcare

TECHNICAL APPROACH

TECHNOLOGIES TO BE USED

Frontend: EJS, HTML5, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, WebSocket

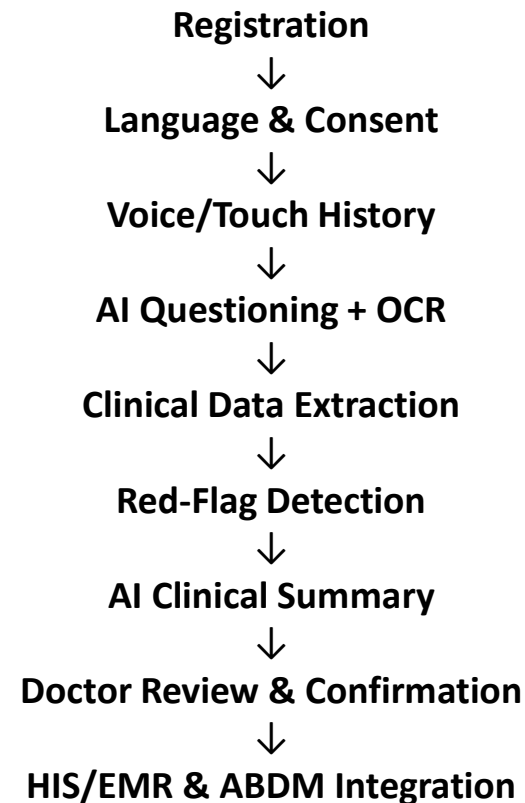
Database: MongoDB

AI/ML: LLM, NLP, Indian Languages, Gemini API

Interoperability: FHIR, ABDM APIs, HIS/EMR Integration

Security: HTTPS/TLS, Encryption, RBAC, Consent & Audit Logs

METHODOLOGY & IMPLEMENTATION PROCESS



Feasibility

Technical: AI/NLP, OCR, Speech-to-Text, EJS, Node.js & MongoDB.

Operational: Simple multilingual kiosk, easy to use.

Integration: API-ready for hospital systems.

Cost: Open-source and scalable.

Scalable: Supports multiple languages and healthcare centres.

Challenges & Solutions

Language: Multilingual AI & continuous training.

AI Accuracy: Doctor-in-the-loop validation.

Privacy: Encryption & access control.

Connectivity: Offline/edge processing.

User Adoption: Simple UI & voice guidance.

Integration: Standardized APIs.

IMPACT AND BENEFITS



Social: Multilingual & accessible healthcare for diverse patients.

Economic: Reduces paperwork, time and operational costs.

Healthcare: Faster, structured and accurate patient history.

Environmental: Reduces paper usage and physical storage.

Scalability: Suitable for clinics, hospitals and rural health centres.

Impact: Faster Care • Better Access • Lower Cost • Paperless Healthcare

WHO – Ethics & Governance of AI in Healthcare — AI safety, privacy, ethics and human oversight.

WHO – AI for Health — Responsible and equitable use of AI in healthcare.

Ayushman Bharat Digital Mission (ABDM) — Digital health records, interoperability and health-data sharing in India.

WHO – AI & Large Multimodal Models — Research on emerging AI applications and risks in healthcare.

Links

[WHO – Ethics & Governance of AI for Health](#)

[WHO – Artificial Intelligence for Health](#)

[Ayushman Bharat Digital Mission \(ABDM\)](#)